

TI 505 and CTI 2500 driver

This module is a driver for CTI 2500-series processors and legacy Siemens/TI 505 PLCs over the CAMP protocol on TCP.

Version 1.0.0 Ignition 8.1.33+ Runs on: any gateway that reaches the PLC network

Contents

1. [1. Overview](#)
2. [2. What You Need](#)
3. [3. Install](#)
4. [4. License](#)
5. [5. Configure](#)
6. [6. Operate](#)
7. [7. Verify](#)
8. [8. Troubleshoot](#)
9. [9. Reference](#)
10. [10. Support](#)

1. Overview

The driver is a standalone OPC UA device type; it does not require any other SparkBridge module.

2. What You Need

- Ignition 8.1.33 or newer.
- A CTI 2572/2500 onboard Ethernet module or compatible 505 gear reachable over TCP, default CAMP port 1505.

3. Install

Install the signed TI 505 / CTI 2500 module through Config > Modules > Install or Upgrade a Module; no build step is required.

4. License

See the README in the bundle for licensing terms.

From release 2.7.0, every SparkBridge module runs on Ignition's standard two-hour module trial (the gateway's Reset Trial button restarts the clock). A verified key overrides the trial entirely and the trial clock is then ignored. Drop the key at <data>/sparkbridge-license.key; each module re-checks the file every 30 seconds, so installing, replacing, or removing a key takes effect with no restart.

5. Configure

Config > OPC UA > Device Connections > Create new Device > TI 505 / CTI 2500.

Setting	Meaning, default
Hostname/IP	PLC or 2572 Ethernet module.
Port	CAMP TCP port, default 1505.
Timeout (ms)	Socket connect/read timeout, default 3000.
Max Read Words	Upper bound per CAMP block read, default 64.
Browse Elements Per Area	Word/discrete elements exposed per area when browsing, default 1024.
Loop/Alarm Browse Count	Loop/alarm numbers exposed per area when browsing, default 64.

Addressing: <area><element>[.<bit>][:<type>[:<len>]]. Discrete areas x y c; word areas v k wx wy stw tcp tcc. Elements are 1-based; bit 1 is the MSB. Examples: v100 (Int16), v100:REAL, v100.3 (bit), v100:STRING:20.

Loop and analog-alarm parameters address by loop/alarm number, for example LERR3 (loop 3 error) or LSP3 (loop 3 setpoint). Confirmed loop codes: LERR, LERRR, LVF, LRSF, LACK. Other loop/alarm codes (LPV LSP LMN LKC LTI LTD LSPH LSPL LHA LLA APV ASP AERR AHA ALA) are provisional: reads work, but writes through the driver are refused (Bad_NotWritable) until independently verified against live hardware.

6. Operate

Day to day, keep provisional loop/alarm codes read-only until they have been checked against a live controller. Writes to a provisional code fail closed with Bad_NotWritable rather than silently succeeding against an unverified mapping.

7. Verify

1. Standalone probe: `java -cp gateway/build/libs/gateway-1.0.0.jar com.cti.ti505.tools.Ti505Probe <host> 1505 V100:REAL.`
2. To confirm a provisional loop/alarm code, read it and compare against the value shown on the HMI or programming software; only promote it to verified after that match.

8. Troubleshoot

Symptom	Cause / fix
Write to a loop/alarm parameter returns Bad_NotWritable	Expected for provisional codes (see the configure section's list); only LERR, LERRR, LVF, LRSF, LACK are confirmed for writes.
Read values look truncated	Check Max Read Words; large blocks may need multiple reads or an increased limit.

9. Reference

Default port	TCP 1505 (CAMP)
Default timeout	3000 ms
Default Max Read Words	64
Confirmed loop codes	LERR, LERRR, LVF, LRSF, LACK

10. Support

Docs page section: </sparkbridge/docs#ti505>. Contact: </sparkbridge/contact>. Release downloads and checksums: [GitHub releases](#).